Tutorial 5

Least-Squares Approximation

Motivation

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- Least squares gives the model that is *closest on average* by minimizing squared error.

Discrete least squares

Discrete vs Continuous Least Squares

Discrete least squares

$$\min_{\alpha} \sum_{i=1}^{n} \big(y_i - g(x_i; \alpha)\big)^2$$

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Used when we have a finite set of measurements.

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Continuous least squares

$$\min_{p \in V} \int_a^b \big(f(x) - p(x)\big)^2 \, dx$$

Used when we approximate a whole function on an interval.

Least Squares vs Interpolation

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Least squares

- used when the data contain measurement error or noise,
- the goal is not to pass through every point,
- the goal is to capture the *overall trend*.

Exercise 1 – Data and Candidate Models

The data are

$$(0.75, 1.2),\ (2, 1.95),\ (3, 2),\ (4, 2.4),\ (6, 2.4),\ (8, 2.7),\ (8.5, 2.6).$$

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The data are

$$(0.75, 1.2),\ (2, 1.95),\ (3, 2),\ (4, 2.4),\ (6, 2.4),\ (8, 2.7),\ (8.5, 2.6).$$

Candidate models given :

$$\begin{aligned}
&\text{(b) } y = 0.1548x + 1.4653\\
&\text{(c) } y = -0.03x^2 + 0.45x + 1\\
&\text{(d) } y = 1.4679e^{0.0793x}\\
&\text{(e) } y = 1.4233x^{0.3114}\\
&\text{(f) } y = \frac{2.928x}{1.081 + x}
\end{aligned}$$

Question: Which approximation looks best by eye?

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Answer: model **(f)**.

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Answer: model **(f)**.
Model (f) follows this shape better than the others. ((c) is not a bad model either !!!) So, by visual inspection:

$$\boxed{\text{Best by eye: model (f)}}$$

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1. Coefficient of determination :measures **how much of the data is explained by the model**

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   Here, $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$ denotes the mean of the observed values, and $\hat{y}_i$ denotes the predicted value corresponding to x_i given by the fitted model.

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$$r_i = y_i - \hat{y}_i.$$

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Interpretation: higher R^2(close to 1), lower RMSE, and residuals with no strong pattern.

Exercise 1 – Question 2: Numerical Comparison

Model	R^2	RMSE
(b) Straight line	0.8028	0.2116
(c) Parabola	0.9373	0.1193
(d) Exponential	0.7283	0.2484
(e) Power law	0.9215	0.1335
(f) Saturated growth	**0.9649**	**0.0893**

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Model	Residuals $r_i = y_i - \hat{y}_i$
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Best model by objective criteria: (f)

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Why "$\approx$" and not "$=$"?

- Usually, the system $A\alpha = y$ has no exact solution.
- This happens because experimental or observed data are often inconsistent or noisy.
- Therefore, we introduce the residual vector

$$r = y - A\alpha,$$

and seek the vector α that makes $||r||$ as small as possible.

From Minimization to the Normal Equations

Expand

$$\phi(\alpha) = \|y - A\alpha\|_2^2 = (y - A\alpha)^T(y - A\alpha).$$

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Differentiate and set the gradient to zero:

$$\nabla\phi(\alpha) = 2A^T A\alpha - 2A^T y = 0.$$

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Hence the **normal equations** are

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Hence the **normal equations** are

$$\boxed{A^T A\alpha = A^T y.}$$

If $\text{rank}(A) = m$, then $A^T A$ is invertible and

$$\boxed{\alpha = (A^T A)^{-1} A^T y.}$$

Small Abstract Example (3×2)

Let

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}, \quad \alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

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Then

$$\phi(\alpha_1, \alpha_2) = \sum_{i=1}^{3} \left(y_i - a_{i1}\alpha_1 - a_{i2}\alpha_2\right)^2.$$

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Differentiate with respect to α_1:

$$\frac{\partial \phi}{\partial \alpha_1} = -2 \sum_{i=1}^{3} a_{i1} \big(y_i - a_{i1}\alpha_1 - a_{i2}\alpha_2\big).$$

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Setting both to zero gives $A^T(y - A\alpha) = 0$, hence $A^T A\alpha = A^T y$.

Exercise 1 – Question 3: Linear Model Setup

We fit

$$y \approx \alpha_0 + \alpha_1 x.$$

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Matrix form:

$$A = \begin{bmatrix} 1 & 0.75 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 6 \\ 1 & 8 \\ 1 & 8.5 \end{bmatrix}, \qquad \alpha = \begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix}, \qquad y = \begin{bmatrix} 1.2 \\ 1.95 \\ 2 \\ 2.4 \\ 2.4 \\ 2.7 \\ 2.6 \end{bmatrix}.$$

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We solve

$$A^T A\alpha = A^T y.$$

Exercise 1 – Question 3: Normal Equations

$$A^T A = \begin{bmatrix} 7 & 32.25 \\ 32.25 & 201.8125 \end{bmatrix}, \qquad A^T y = \begin{bmatrix} 15.25 \\ 78.5 \end{bmatrix}.$$

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Solving gives

$$\alpha_0 = 1.465322, \qquad \alpha_1 = 0.154814.$$

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Solving gives

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Hence

$$\boxed{y = 1.4653 + 0.1548x.}$$

Exercise 1 – Question 4: Quadratic Model Setup

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Matrix form:

$$A = \begin{bmatrix} 1 & 0.75 & 0.75^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 6 & 6^2 \\ 1 & 8 & 8^2 \\ 1 & 8.5 & 8.5^2 \end{bmatrix}, \quad \alpha = \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{bmatrix}.$$

Exercise 1 – Question 4: Quadratic Model Setup

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Again we solve

$$A^T A\alpha = A^T y.$$

Exercise 1 – Question 4: Normal Equations

$$A^T A = \begin{bmatrix} 7 & 32.25 & 201.8125 \\ 32.25 & 201.8125 & 1441.546875 \\ 201.8125 & 1441.546875 & 10965.37890625 \end{bmatrix},$$

Exercise 1 – Question 4: Normal Equations

$$A^T A = \begin{bmatrix} 7 & 32.25 & 201.8125 \\ 32.25 & 201.8125 & 1441.546875 \\ 201.8125 & 1441.546875 & 10965.37890625 \end{bmatrix},$$

$$A^T y = \begin{bmatrix} 15.25 \\ 78.5 \\ 511.925 \end{bmatrix}.$$

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Solving yields

$$\alpha_0 = 0.990728, \qquad \alpha_1 = 0.449901, \qquad \alpha_2 = -0.030694.$$

Exercise 1 – Question 4: Normal Equations

$$A^T A = \begin{bmatrix} 7 & 32.25 & 201.8125 \\ 32.25 & 201.8125 & 1441.546875 \\ 201.8125 & 1441.546875 & 10965.37890625 \end{bmatrix},$$

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Solving yields

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Hence

$$\boxed{y = 0.9907 + 0.4499x - 0.03069x^2.}$$

Exercise 2 – Model Setup

We seek the best linear model

$$g = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2$$

for the data:

$$(5, 3, 0.8), \ (3, 4, 0.8), \ (1, 5, 0.6), \ (2, 1, 0.4).$$

Exercise 2 – Model Setup

We seek the best linear model

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for the data:

$$(5, 3, 0.8), \ (3, 4, 0.8), \ (1, 5, 0.6), \ (2, 1, 0.4).$$

Write

$$A\alpha \approx y,$$

with

$$A = \begin{bmatrix} 1 & 5 & 3 \\ 1 & 3 & 4 \\ 1 & 1 & 5 \\ 1 & 2 & 1 \end{bmatrix}, \qquad \alpha = \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{bmatrix}, \qquad y = \begin{bmatrix} 0.8 \\ 0.8 \\ 0.6 \\ 0.4 \end{bmatrix}.$$

Exercise 2 – Solve the Normal Equations

$$A^T A = \begin{bmatrix} 4 & 11 & 13 \\ 11 & 39 & 34 \\ 13 & 34 & 51 \end{bmatrix}, \qquad A^T y = \begin{bmatrix} 2.6 \\ 7.8 \\ 9 \end{bmatrix}.$$

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Solving gives

$$\alpha_0 = \frac{29}{210}, \qquad \alpha_1 = \frac{19}{210}, \qquad \alpha_2 = \frac{17}{210}.$$

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So

$$\begin{bmatrix} 4 & 11 & 13 \\ 11 & 39 & 34 \\ 13 & 34 & 51 \end{bmatrix} \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} 2.6 \\ 7.8 \\ 9 \end{bmatrix}.$$

Solving gives

$$\alpha_0 = \frac{29}{210}, \qquad \alpha_1 = \frac{19}{210}, \qquad \alpha_2 = \frac{17}{210}.$$

Therefore

$$\boxed{g(x_1, x_2) = 0.1381 + 0.0905 x_1 + 0.0810 x_2.}$$

Exercise 3 – Linear Trend Setup

We model sales by a linear trend

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We again solve the least-squares problem

$$A^T A\alpha = A^T y.$$

Exercise 3 – Solve the Normal Equations

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \end{bmatrix}, \quad \alpha = \begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix}, \quad y = \begin{bmatrix} 3000 \\ 3200 \\ 3250 \\ 3450 \\ 3800 \\ 3850 \\ 4100 \end{bmatrix}.$$

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$$\boxed{y \approx 2969.64 + 183.93x.}$$

Exercise 3 – Forecast for Year $N+1$

Year $N+1$ corresponds to $x = 7$.

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So the forecast is

$$\boxed{\text{Estimated sales revenue for year } N + 1 \approx 4257.}$$

Exercise 4 – Continuous Least Squares Setup

We want the polynomial of degree at most 1

$$L_1(x) = \alpha_0 + \alpha_1 x$$

that best approximates

$$f(x) = e^x$$

on $[0, 1]$ in the continuous least-squares sense.

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The continuous normal equations are

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$$\int_0^1 x\left(e^x - \alpha_0 - \alpha_1 x\right) dx = 0.$$

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Solving gives

$$\alpha_1 = 18 - 6e, \qquad \alpha_0 = 4e - 10.$$

Exercise 4 – Final Polynomial

Therefore

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